

Traffic Classification of User Behaviors in Tor, I2P, ZeroNet, Freenet

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Abstract—In recent years, more and more anonymous network have been developed. Since user's identity is difficult to trace in anonymous networks, many illegal activities are carried out in darknet. In this paper, we propose a hierarchical classifier of darknet traffic which can distinguish four types of darknet(Tor, I2P, ZeroNet, Freenet) and 25 darknet users' behavior. Due to the lack of public datasets, we deployed a darknet data probe that can capture real darknet traffic in Tor, I2P, ZeroNet, Freenet. After collecting and labeling darknet traffic, we extract 26 time-based flow features that can represent the characteristics of darknet traffic and train a hierarchical classifier constructed by 6 local classifiers. Results show that the classifier can easily distinguish Tor, I2P, ZeroNet, Freenet four kinds of darknet clients with an accuracy of 96.9% and identify 8 kinds of user behaviors for each type of darknet with an accuracy of 91.6% on average. With the help of this hierarchical classification method, darknet user behaviors can be accurately distinguished at the traffic exit.

Keywords—Darknet, Traffic Classification, User Behavior, Hierarchical Classification, Tor, I2P, Freenet, ZeroNet

I. INTRODUCTION

Darknet[1] is defined as a kind of network that can only be accessed through special configuration, software, authorization, or non-standard communication protocols and ports. Addresses of darknet users and the content of communication are usually anonymous and hard to trace back. As various darknet become more and more mature, darknet has gradually become a breeding ground for terrorism and crimes[2]. The anonymity of the darknet makes it a trading platform for political intelligence and personal privacy data. From the perspective of network administrators, in order to better supervise or even block the possible illegal activities in darknet, the capability of classifying users' behavior in darknet is extremely needed.

Among the various darknet developed in recent years, The Onion Router(Tor) [3], Invisible Internet Project(I2P) [4], ZeroNet [5], Freenet [6] are the most popular. In the traffic classification of darknet, many studies only focused on a single type of darknet, among which the most researched is Tor [7][8]. I2P [9] and Freenet [10] are also discussed in a similar way. However, these studies did not go deep into the level of user behaviors. Montieri et al [11][12] have studied the classification of three types of darknet(Tor, I2P, and JonDonym), and the applications in the three types of darknet. But it only covered a small range of applications, and performed not well enough(with an accuracy of 66.76%). It can be seen that darknet traffic classification is a new research

field that worth further study. According to our knowledge, there are few studies on the analysis of user behavior in darknet, especially for the new type of darknet(like ZeroNet) and new applications in mature darknets. For this reason, we conducted research on darknet traffic classification, and refined the granularity of classification to the level of user behavior.

The foundation of darknet traffic research lies in dataset, the lack of public datasets is also a challenge for study. Many previous works[13] are based on dataset captured in simulation environments which can not represent the traffic in the real world. However, the author of the dataset generated in real environments[14] often refuses to make it public due to privacy considerations. To overcome this challenge, we deployed traffic collecting probes in the real darknet environment to capture and label user behaviors traffic by ourselves. After researching on the possible users behaviors in Tor, I2P, ZeroNet, Freenet, we divide user behaviors into 8 categories: Browsing, Chat, E-mail, Audio-streaming, Video-streaming, File Transfer, P2P, VoIP. The dataset we used in experiment has been published on the web after anonymization.

When the traffic classification is extended to the level of user behaviors for 4 types of darknet, the number of classification categories becomes a new problem for training (In this paper, there is 25 types of user behaviors). In order to improve the accuracy of identification and avoid the mistake of recognizing Tor's File Transfer behavior as Freenet's Web Browsing behavior, we purposed a hierarchical classification method with three layers. The hierarchical classifier we trained first determines whether it is darknet traffic, then determines the type of darknet, and finally predicts what user behavior is. The accuracy of darknet traffic classification is 99.42%, and the accuracy of darknet type classification is 96.9%, while the accuracy of user behavior classification is 91.6% on average. The experiment show that the hierarchical classifier performs better than the best flat classifier.

In the procedure of training local classifiers of the hierarchical classifier, we considered six machine learning algorithms(LR, DT, RF, GBDT, XGBoost, LightGBM) and two deep learning algorithms(MLP, LSTM) to test which algorithm is more suitable for the darknet traffic scenario. The results show that ML classifiers generally perform better than DL classifiers when the feature extraction can well represent the characteristics of traffic (In this paper, we extract 26 time-based flow feature from pcap for training). Among them, XGBoost algorithm performs the best, which is also used as

the best flat classifier to be compared with our hierarchical classifier.

Contribution. The main contributions of this paper are as follows:

- 1) Due to the lack of public darknet datasets, we deployed traffic collecting probes in the real darknet environment to capture and label 8 kinds of user behavior traffic(Browsing, Chat, E-mail, Audio-streaming, Video-streaming, File Transfer, P2P, VoIP) in Tor, I2P, ZeroNet, Freenet and published the dataset on the web¹.
- 2) We propose a three-layer hierarchical classification method for Tor, I2P, ZeroNet, Freenet traffic, which can refine the granularity to the level of user behavior. Compared with the best flat classifier, hierarchical classifier has higher accuracy in the scenario with large number of classification categories(In this paper, there is 25 types of user behaviors).
- 3) We considered six machine learning classifiers(LR, DT, RF, GBDT, XGBoost, LightGBM) and two deep learning classifiers(MLP, LSTM) for the training of local classifiers. The results show that the XGBoost algorithm performs best in this darknet scenario. DL algorithm does not perform well when the feature extraction can well represent the characteristics of traffic. The hierarchical classifier constructed by 6 local classifiers has an accuracy of 96.9% in darknet type classification, and an accuracy of 91.6% on average in user behavior classification.

The rest of this paper is organized as follows. Section II discusses related works, whereas Section III introduces the generation of datasets including the collection of traffic and flow features extraction. Section IV describes the algorithms used in training our hierarchical classifier. In Section V we present the experimental results of 6 local classifiers under 8 algorithms and the comparison between our hierarchical classifier and the best flat classifier. Finally, in Section VI, the conclusion and future work of this research are outlined.

II. RELATED WORK

Traffic Classification is a very important part of network defense, supervision and anomaly detection, so it has been the focus of security researchers for many years. JV Gomes[15] presented a survey that described existing traffic classification approaches, including port-based, DPI-based and flow-based methods. In this paper, we will focus on the traffic classification in darknet scenarios. As darknet traffic is only a drop of the bucket compared to Internet traffic, darknet researchers first pay attention to how to classify darknet traffic from the complex Internet traffic. Hodo Elike et al. [16] proposed a solution to the classification problem of Tor traffic and nonTor traffic. In feature extraction, ISCXFlowMeter [17] is used to extract time-based features and flow features. Two algorithms, ANN and SVM, are selected for experimental testing, and the results show a high accuracy. Barker et al.

[18] collected normal HTTPS traffic, Tor HTTPS traffic to design an unsupervised classifier based on the length of the packet. He Yongzhong et al. [19] adopted a feature matching classification method based on SVM for the meek-based Tor anonymous traffic. Some typical static and dynamic traffic features are extracted, including sequence of sending and receiving, length entropy and length variation.

In recent years, new darknet like I2P and Freenet have gradually become another option for anonymous communication. Antonio Montieri et al. [11][12] have studied the traffic classification of different anonymous communication networks. In these two papers, they designed a classification for Tor, I2P, and JonDonym and compared four algorithms Naïve Bayes, Bayesian Network, C4.5, Random Forest.

Identifying user behavior on darknet plays an important role in darknet traceability and management. He Gaofeng et al. [20] established HMM model (Hidden Markov Models) to identify four types of user behaviors in darknet: P2P, FTP, IM, and Web. Rob Jansen et al. [21] proposed the Web Fingerprinting (WF) method by deploying it on the Tor middle relay. This method can accurately identify which Web site the user visits.

It can be seen that most of the research on darknet traffic focus on a single type of darknet like Tor, less care about some nearly-emerged darknet like ZeroNet. In addition, there are few studies go deep into the user behavior in darknet. Most of them only focus on whether it is darknet traffic, and the dataset is often kept in secret which brings difficulties to further research. In the view of these reason, we conduct in-depth research on the issue of user behavior in Tor, I2P, Freenet and the emerging darknet, ZeroNet. We collect and publish the darknet user behavior dataset, and refines the granularity of darknet traffic identification to user behavior so as to advance related work.

III. DATASET GENERATION

In this section, we will introduce the generation of dataset that used in our experiment, including the procedure of traffic collection and flow feature extraction which can convert continuous packets data into discrete feature vectors.

A. Darknet Traffic Collection

One of the contributions of this paper is the Tor, I2P, ZeroNet, Freenet traffic datasets that we used in our experiments. To generate dataset that contains eight types of user behaviors traffic(Browsing, Chat, E-mail, Audio-streaming, Video-streaming, File Transfer, P2P, VoIP) in four darknets, we built four darknet traffic probes and created accounts for each darknet to use specific applications. Following, we give a detailed description of eighth types of user behaviors traffic.

Browsing: Under this label, We simulate web browsing behaviors in Tor, I2P, ZeroNet, Freenet. We visit DuckDuckGo, Twitter and other forums, social media, and some hidden sites(domains ending in .onion) by Tor proxy; I2P forums(i2pforum.i2p), documents(echelon.i2p) and news communities(planet.i2p) with I2P http proxy.index sites(Enzo's index,ZeroSites), forum(ZeroTalk), blogs(ZeroBlog) in Freenet and ZeroNet.

¹<https://github.com/huyz97/darknet-dataset-2020>

TABLE I: Darknet Dataset at User Behavior Granularity

Darknet Type	User Behavior								
	Browsing	Chat	Email	File Transfer	P2P	Audio	Video	VoIP	Total
Tor	1281	841	553	1077	1018	1567	1703	592	8632
I2P	1921	442	1084	1791	2910	-	-	-	8148
ZeroNet	7972	1531	352	2157	1394	820	1251	-	15477
Freenet	4990	1123	2980	4897	-	-	2397	-	16387

Email: In Tor, emails through SMTP/S, POP3/SSL, IMAP/SSL are forwarded through socks5 Tor proxy. In I2P, we send and receive emails by susimail(an I2P email application) after configuring the smtp server(smtp.postman.i2p:25) and pop3 server(pop3.postman.i2p:110). Emails in Freenet are only supported under IMAP protocol by Freemail which listens 127.0.0.1:4025 by default. Emails in ZeroNet are end-to-end encrypted emails by ZeroMail and ZeroID is needed to used as user identity.

Chat: This label represents the instant-messaging behavior including text, picture, file. In Tor, we use ICQ and Skype to simulate the chat behavior between users by using Tor socks5 proxy. In I2P, after enabling SAM protocol bridge, I2P will listen port 7656 on localhost by default. I2P-Messenger and I2Ptalk, open source chat application in I2P, can be used for peer-to-peer chat. FMS(Freenet Message System) is a popular chat application on Freenet in which users take their SSK as identity. As for ZeroNet, there are many chat rooms in different languages instead of peer-to-peer chat behavior since ZeroNet is developed based on BitTorrent protocol.

File Transfer: This label represents the user behavior of downloading and uploading file, which is also the focus in darknet supervision. Within this label we captured FTP traffic using socks5 proxy in Tor. For the other three darknets, we visit muwire.i2p for I2P traffic collection, IFS site for ZeroNet, and Fuqid application for Freenet.

P2P: This label is used to identify file-sharing protocols. Vuze application is used for finding peers and downloading files through Tor socks5 proxy. As for I2P, we use I2Psnark as anonymous BitTorrent client to download shared-file in I2P which can be obtained from PaTracker site.

Audio-streaming: Audio label represents a continuous and steady stream of audio data. We captured this type of traffic from Spotify in Tor, and from IFS site in ZeroNet.

Video-streaming: This label identifies a continuous stream of video data. We captured traffic from Youtube by Tor socks5 proxy and ZeroNet traffic from Zerotube site.

VoIP: The voice over IP label represents the traffic generated by voice applications which is only available in Tor. We captured phone-call traffic by using Skype through Tor proxy.

B. Flow Features Extraction

After finishing the data collection, it is necessary to extract the characteristics of the traffic. The general definition of flow is often a five tuple {source IP, destination IP, source port, destination port, communication protocol}. But in complex

network scenarios like darknet, it is hard to distinguish different flow only relying on these 5 feature dimensions. We extracted 26 features from the darknet traffic with the help of CICFlowMeter [17]. These traffic features are extracted in both directions where the first arrived packet determines the forward and backward directions. A detailed description of 26 features we used in experiment is as follows:

Flow Duration: The duration of the flow. TCP packets use the FIN flag packet as the end of the flow, while the UDP packets use the flow timeout as the end mark. The timeout period in this paper is 12 seconds.

Total Fwd/Bwd Packets: The number of packets in forward/backward direction.

Packet Length : The length of the packet (Min/Mean).

Fwd/Bwd/Flow IAT: The value of Inter Arrival Time between two packets in forward/backward/both direction (Total/Min/Max/Mean).

Flow Packets/s: The number of flow packets per second.

Flow Bytes/s: The Byte rate of flow packets per second.

Idle: The amount of time that a flow was idle before becoming active (Min/Max/Mean/Std).

Fwd/Bwd Init Win Bytes: The init window size in forward/backward direction.

ACK Flag Count: The count of ACK packet in TCP flows.

Hence one can see that except the easily forged elements such as port and IP, these 26 time-based flow features can well represent the characteristics of the traffic flow. After feature extraction, the statistical results of the dataset are shown in Table 1.

IV. CLASSIFICATION ALGORITHMS

In this section, we will introduce the hierarchical classification method used in the experiment, and review 6 ML algorithms and 2 DL algorithms, which are compared in the selection of local classifiers.

A. Hierarchical Classification

Since each darknet has 5 to 8 user behaviors, directly training a multi-class classifier will result in too many categories and lower accuracy. In this scenario, we propose a hierarchical classification method as darknet traffic identification model. The hierarchical classifier structure is shown in Fig.1.

The darknet traffic classifier has three layers. First, determine whether it is darknet traffic or normal internet traffic. If it is darknet traffic, continue to identify what darknet type it is, and then determine what user behavior is conducted. The advantage of using a hierarchical classifier is that the

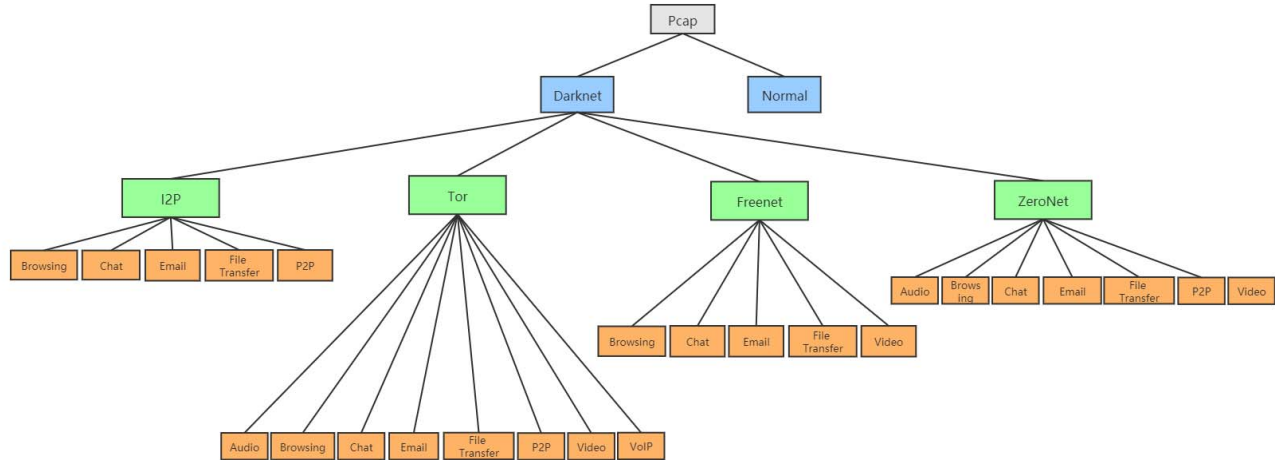


Fig. 1: Hierarchical classifier structure.

local classifier of each parent node can be trained separately, and different classification algorithms can be selected for each local classifier, which is more flexible for training. The disadvantage of this method is that there may be error propagation. Once an error occurs in the classification of one layer, it will directly affect the final classification result.

The hierarchical classifier is composed of the local classifiers of each sub-task, and there are three training methods:

- 1) Training a binary classifier for each node, similar to one-versus-rest method, has a large performance overhead.
- 2) Train a multi-classifier for each layer. In the darknet traffic scenario, only three classifiers need to be trained, which has low overhead and poor accuracy. And because the behaviors of different types of dark web users are not similar, it is not applicable.
- 3) Train a multi-classifier for each parent node, and each classifier is independent and judged according to a tree structure. This training method is relatively intuitive and has a compromised performance.

In the darknet traffic scenario of this paper, the third type of hierarchical classifier training method is adopted, which means 6 independent classifiers need to be trained separately. Since each classifier is independent, each parent node can choose the best performance Classification algorithm. Therefore, in order to achieve the best hierarchical classification effect, it is necessary to select the best local classifier for each parent node.

B. Local Classifier

We compared the result of six ML classifiers(LR, DT, RF, GBDT, XGBoost, LightGBM) which is commonly used in traffic classification[22][23] and two basic DL classifiers(MLP, LSTM) as local classifiers in this darknet scenario.

Logistic Regression (LR): It belongs to supervised learning which is often used for binary classifications. It can also be applied to multi-class scenarios according to the methods of

OvR (one-vs-rest) and MvM (many-vs-many). The essence of logistic regression is to assume that the data obey the distribution, and then use maximum likelihood estimation to estimate the parameters. In this experiment, OvR method is adopted, class_weight parameter is balanced, which means the weight of each categories will be considered, cross-validation parameter is 5 and the maximum number of iterations is 1000. The remaining parameters are set to default values.

Decision Tree (DT): It is a non-parameter supervised learning method used for classification and regression. Its purpose is to create a model to learn simple decision rules from data features to predict the value of a target variable. The decision tree can handle the problem of multi-classification well, but it is also easy to produce an overly complex model, which leads to poor generalization performance of the model and causes over-fitting. By controlling the maximum depth of the decision tree and the number of samples of leaf nodes, the occurrence of overfitting can be prevented. Decision tree algorithms include ID3, C4.5, C5.0 and CART. In this paper we use the tuned CART tree algorithm in Sklearn [24], and the remaining parameters are set to default values.

Random Forest (RF):[25] A type of ensemble learning that combines the prediction results of the multiple estimators to obtain better generalization. The randomness of sampling and node splitting can reduce the variance of the DT and avoid overfitting. The randomness of the random forest construction process can produce a set of different decision trees. By taking the average of each decision tree, better results can be obtained. In this paper, the number of estimators is 100, the maximum feature number is 5, and the sampling method is bootstrap sample. The remaining parameters are set to default values.

Gradient Boosting Decision Tree (GBDT): A type of ensemble learning. GBDT is an iterative decision tree algorithm composed of multiple decision trees. The core of the GBDT is to use the value of the negative gradient of the loss function

in the current model as approximations of the residual. In this paper, we use the GBDT classifier implemented in sklearn, and the number of estimators is 100.

Extreme Gradient Boosting (XGBoost):[26] The XGBoost method published by the author Tianqi Chen at the KDD conference is essentially a Boosting algorithm. Compared with the GBDT implemented in sklearn, XGBoost can be regarded as an upgraded version. The boosting strategies of the two are similar, but the fitting direction of XGBoost is more accurate and faster. In this paper, the number of estimators of XGBoost is 100.

LightGBM:[27] It is also a gradient boosting algorithm based on decision trees proposed by Microsoft. Compared with XGBoost and traditional GBDT algorithms, LightGBM uses histogram algorithm to replace Pre-Sorted algorithm in finding the optimal partition node, which consumes lower memory and

faster data segmentation. Different from the level-wise growth strategy adopted by XGBoost, LightGBM adopts a leaf-wise growth strategy which can grow a deeper decision tree.

Multilayer Perceptron (MLP): One of the artificial neural networks. By mapping a set of input vectors to output vectors, each layer is fully connected to the next layer. Each node in MLP is a non-linear activation function neuron and each connection is equipped with a weight. The MLP contains at least one hidden layer in addition to the input layer and output layer. In this paper, the number of hidden layers is 10, each layer has 128 neurons, the optimization algorithm is Adam, the activation function is softmax, the loss function is binary_crossentropy, batch_size is 100, and the number of iterations is 100.

Long short-term memory (LSTM):[28] It is a special recurrent neural network (RNN), mainly to solve the problem

TABLE II: Accuracy and F1-score of 6 local classifier in darknet hierarchical classification under 8 algorithms

Model	Local Classifiers											
	Darknet-Normal		Darknet-type		Tor-behavior		I2P-behavior		Freenet-behavior		ZeroNet-behavior	
	accuracy	F1-score	accuracy	F1-score	accuracy	F1-score	accuracy	F1-score	accuracy	F1-score	accuracy	F1-score
LR	93.71%	94.26%	62.80%	62.89%	52.26%	51.89%	33.91%	29.20%	41.79%	36.42%	51.42%	48.11%
DT	99.30%	99.33%	96.16%	96.16%	83.79%	83.82%	90.51%	90.52%	96.66%	96.66%	90.63%	90.65%
RF	99.38%	99.41%	96.55%	96.55%	84.64%	84.65%	90.76%	90.76%	96.66%	96.67%	91.23%	91.18%
GBDT	98.51%	98.58%	95.23%	95.23%	84.29%	84.27%	90.84%	90.82%	96.85%	96.85%	86.67%	86.16%
XGBoost	99.42%	99.45%	96.85%	96.85%	85.35%	85.38%	91.29%	91.28%	96.75%	96.74%	92.98%	92.98%
LightGBM	99.34%	99.37%	96.33%	96.33%	85.24%	85.28%	90.27%	90.25%	96.46%	96.46%	93.17%	93.18%
MLP	-	-	94.27%	88.09%	94.82%	71.13%	-	-	-	-	92.02%	65.05%
LSTM	-	-	87.18%	87.17%	73.04%	73.48%	-	-	-	-	66.31%	63.76%

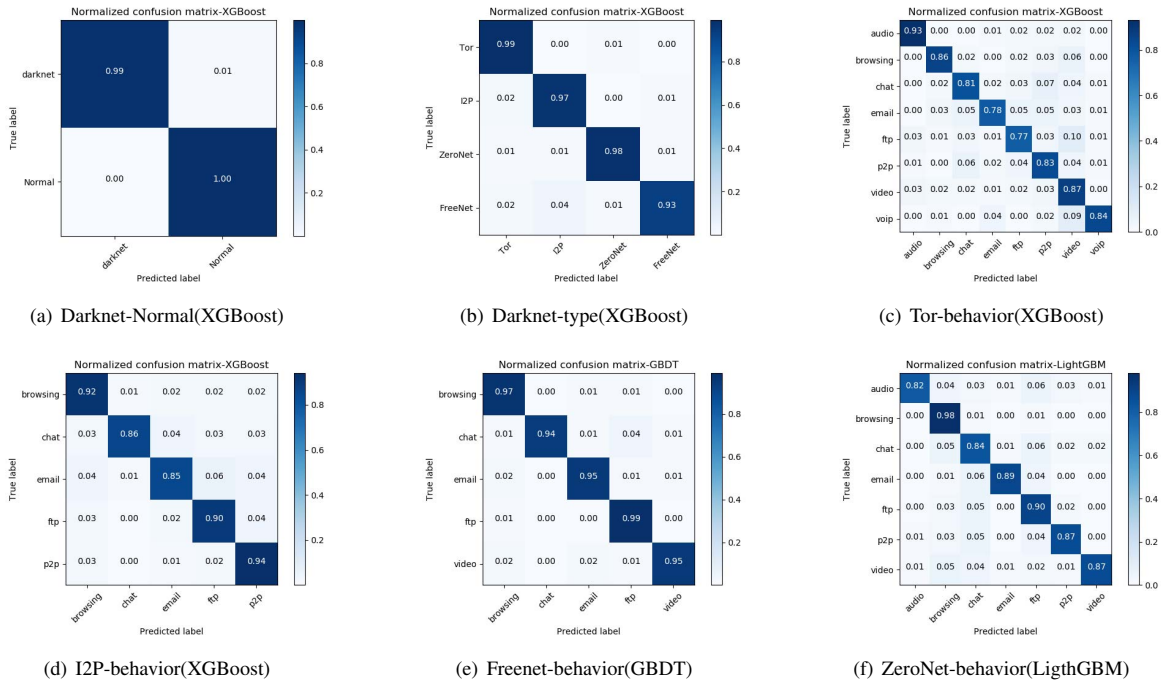


Fig. 2: Confusion Matrix for 6 Local Classifiers

of gradient disappearance and gradient explosion during long sequence training. Compared to ordinary RNNs, LSTM can perform better in longer sequences. In this paper, the number of hidden layers is 1, each layer has 128 neurons, the activation function is softmax, the optimization algorithm is rmsprop, batch_size is 64, and the number of iterations is 100.

We trained each parent node of hierarchical classifier separately and select the best performed algorithm for local classifier. Performance comparison results will be introduced in section V.

V. EXPERIMENTS AND RESULTS

In this section, we will introduce the performance comparison of 6 local classifiers with different algorithms, and then introduce the user behaviors identifying performance of the hierarchical classifiers we proposed. We compare hierarchical classification method with flat classification method to prove the improvement.

A. Performance Evaluation Metrics

We used accuracy, precision, recall, F1-score and confusion matrix to evaluate the performance of classifiers.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

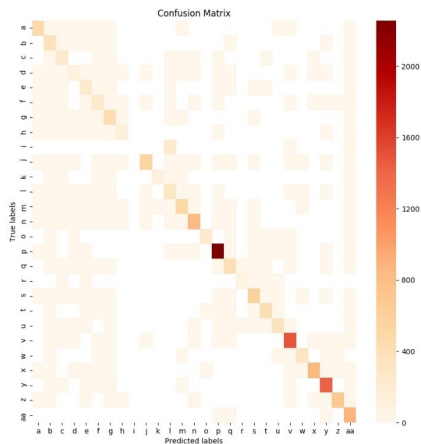
B. Local Classifier Performance

In order to choose the most suitable classification algorithm for each local classifier, we use the dataset collected in Section III as darknet traffic and dataset from Stratosphere Lab [29] as normal internet traffic. 6 local classifiers are Darknet-Normal, Darknet-type, Tor-behavior, I2P-behavior, Freenet-behavior, ZeroNet-behavior. For the first two classifiers, we split train-test dataset with corresponding percentages 70%-30%, while the last four classifiers use 80%-20% train-test set splitting. We take accuracy and F1-score as the criteria. The performance results are shown in Table II, where the bold is the best performed one in local classifiers.

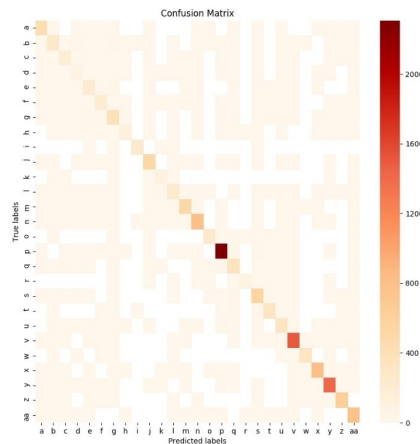
According to the performance results of each local classifier, we found that in the darknet traffic scenario, the ML method generally performs well. By contrast, DL method has a higher training cost, and the performance is not as good as the ML method when the feature extraction can well represent the traffic. We finally choose XGBoost for Darknet-Normal classifier, Darknet-type classifier, Tor-behavior classifier, I2P-behavior classifier, GBDT for Freenet-behavior classifier and LightGBM for ZeroNet-behavior classifier. Fig.2 shows the confusion matrix of 6 local classifiers.

C. Hierarchical Classifier vs. Flat Classifier

The darknet traffic hierarchical classifier has 25 classes of different user behavior. we compared our Hierarchical Classifier with the Best Flat Classifier in darknet user behavior identification with a train-test dataset splitting in 70%-30%. We take XGBoost model as the flat classifier since it performed well in this scenario. The accuracy, precision, recall, F1-score of the hierarchical classifier and the flat classifier in 25 types of user behavior on the test dataset are shown in Table III. The confusion matrix of the hierarchical classifier and flat classifier on the test dataset is shown in Fig.3. Label



(a) Confusion Matrix of Hierarchical Classifier



(b) Confusion Matrix of Flat Classifier

Fig. 3: Confusion Matrix

TABLE III: Performance Comparison of Hierarchical Classifier and Flat Classifier of Darknet user behaviors

Model	Accuracy	Precision	Recall	F1-score
Hierarchical Classifier	92.46%	93.25%	92.47%	92.06%
Flat Classifier	88.57%	88.66%	88.57%	88.35%

a to aa in the matrix represent user behavior such as tor-browsing, i2p-chat. From the confusion matrix, it can be seen that the false predicted rate of flat classifier is significantly higher than that of the hierarchical classifier. It prove that our hierarchical classifier has a better recognition effect in this darknet scene, and there are few cases of error propagation. Thus this classification method has a certain capability of detecting darknet user behaviors.

VI. CONCLUSION

This paper researches on the traffic classification of user behavior in Tor, I2P, ZeroNet, Freenet. According to our knowledge, there are few studies on the analysis of darknet user behaviors, especially for the new type of darknet like ZeroNet and new applications in mature darknets. We propose a hierarchical classification method for the identification of darknet user behavior which is more suitable in this darknet scenario than flat classification method since there is a large number of predicted categories. The experiment shows that this method can recognize 4 types of darknet with an accuracy of 96.9% and recognize 25 user behaviors with an accuracy of 92.46%. Since the previous research did not give out public traffic datasets on the granularity of user behavior, we deployed a traffic collection probe in real darknet environment and published the dataset we used in the experiment. In the experiment, we extract 26 time-based flow feature from traffic file to train 6 local classifiers which constitute the three-layer hierarchical classifier. The result also shows that the DL method does not perform well when the feature extraction can accurately represent traffic characteristics. In the future, we plan to further supplement the traffic under multiple network bandwidths to expand the darknet dataset, which can be useful for improving the generalization of the model. We also try to link it with IDS to provide a supervision method that detects and blocks dangerous behaviors on the darknet.

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